

# Ramratan Padhy

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## EDUCATION

### SRM University AP

*Bachelor of Technology in Computer Science Engineering*

- Year 1 | SGPA: 9.0/10.0

Amaravati, Andhra Pradesh

2025 – Present

## EXPERIENCE

### Next Tech Lab AP

*Associate*

Amaravati, Andhra Pradesh

Oct 2025 – Present

- Research associate at a QS ReImagine awarded student society.
- Collaborating with a research collective to explore low-level implementations of emerging technologies.
- Contributing to peer-to-peer technical workshops on systems programming and AI architecture.

### Notecove (Stealth Startup)

*Technical Content Writer*

Remote

May 2025 – Aug 2025

- Authored 30+ comprehensive technical articles and guides on emerging software trends and AI ethics.
- Synthesized complex engineering concepts into accessible documentation for a specialized audience.
- Collaborated with the founding team to define the technical voice and content strategy for the platform.

### Hush (Startup)

*ML Intern*

Remote

March 2026 – Present

- Worked with image models to discern user profiles to keep the app authenticity.
- Collaborated with the founding team to define and enhance user experience.

## PROJECTS

### OpenNote | Python, Harness Engineering

In-Progress

- An open source, privacy-focused agentic harness to help you study.
- Made a harness to function as an educational tool, that can create various artifacts like mind-maps, videos, audios.
- Supports various LLMs, including locally run ones.
- Features native terminal support

### Grand Tribunal | Python, Modal, Nemotron, VoiceCPM

Completed

- Shipped an Ace Attorney-style turn-based AI debate game utilizing voice input for an immersive, hands-free player experience. Trained and deployed two specialized Nemotron models on character personality datasets to serve as dynamic opponents and logical judges, interfaced through a custom Gradio frontend.

### Argus | Python, Gemma 4, IoT Simulation

Completed

- Simulated a drone swarm that collectively maps any terrain using Gemma 4 31B vision via Cerebras.
- Engineered autonomous agent drones sharing a collective memory that aggregates into a unified semantic map displayed on a CesiumJS 3D globe.
- Used Gemma 4 inference to classify areas on the terrain, map traffic congregation spots
- Worked with FastAPI to serve as the backend for the simulation

### MinimalMe | HTML, CSS, Web Optimization

Completed

- Architected a zero-dependency, single-file portfolio site optimized for instant edge delivery, achieving sub-300ms load times and a perfect 100/100 Lighthouse performance score.
- Engineered a completely static, JavaScript-free architecture that inherently eliminates third-party vulnerabilities, maximizing security while ensuring optimal crawlability for modern AEO and SEO bots.

### AXtheR (Windows) | C, Windows API, Wintun, Cryptography

Completed

- Developed a high-performance, zero-dependency, easy to run UDP VPN for Windows by integrating the Wintun kernel driver and Winsock2.
- Engineered a zero-heap allocation data path with multi-threaded asynchronous I/O and dynamic Windows routing table manipulation.
- Implemented a self-contained, RFC-compliant ChaCha20-Poly1305 cryptographic engine from scratch in C with constant-time MAC verification.

### Independent Research on BitNet-1.58b | Python, Machine Learning, Mathematics

Completed

- Conducting novel research and experimentation on BitNet-1.58b by altering its context length.

## TECHNICAL SKILLS

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**Languages:** Python, C, LaTeX, Godot (GDScript)

**Developer Tools:** APIs, Vector Databases, PyTorch, Scikit-Learn, TensorFlow, Windows API, Antigravity

**Core Concepts:** Data Structures, Transformer Architectures, LLaMA Architecture, RAG, Technical Documentation, Mechanistic Interpretability, Model Evaluation

**Soft Skills:** Technical Communication, Research Synthesis, Problem Solving, Teamwork